QUESTION 1

```
import pandas as pd

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\players_info -
Players.csv")

print("First Five Records")
print(df.head())

print("\nColumn Names")
print(df.columns)

total_players = len(df)
print("\nTotal Number of Players =", total_players)

print("\na) Probability Distribution of Gender")

gender_frequency = df["gender"].value_counts()
gender_probability = gender_frequency / total_players

gender_result = pd.DataFrame({
    "Frequency": gender_frequency,
    "Probability": gender_probability,
    "Percentage": gender_probability * 100
})

print(gender_result)
print("\nTotal Probability =", gender_probability.sum())

print("\nb) Probability Distribution of Batting Style")

batting_frequency = df["batting_style"].value_counts()
batting_probability = batting_frequency / total_players

batting_result = pd.DataFrame({
    "Frequency": batting_frequency,
    "Probability": batting_probability,
    "Percentage": batting_probability * 100
})

print(batting_result)
print("\nTotal Probability =", batting_probability.sum())

print("\nc) Probability Distribution of Bowling Style")

bowling_frequency = df["bowling_style"].value_counts()
bowling_probability = bowling_frequency / total_players

bowling_result = pd.DataFrame({
    "Frequency": bowling_frequency,
    "Probability": bowling_probability,
    "Percentage": bowling_probability * 100
})

print(bowling_result)
print("\nTotal Probability =", bowling_probability.sum())

print("\nd) Probability Distribution of Player Position")

position_frequency = df["position"].value_counts()
position_probability = position_frequency / total_players

position_result = pd.DataFrame({
    "Frequency": position_frequency,
    "Probability": position_probability,
    "Percentage": position_probability * 100
})

print(position_result)
print("\nTotal Probability =", position_probability.sum())

print("\ne) Probability Distribution of Country")

country_frequency = df["country"].value_counts()
country_probability = country_frequency / total_players

country_result = pd.DataFrame({
    "Frequency": country_frequency,
    "Probability": country_probability,
    "Percentage": country_probability * 100
})

print(country_result)
print("\nTotal Probability =", country_probability.sum())
```

QUESTION 2

```
from math import comb
import pandas as pd
import matplotlib.pyplot as plt

n = 10

p = 0.80

q = 1 - p

x = 10
prob_everyone = comb(n, x) * (p ** x) * (q ** (n - x))

print("QUESTION 1")
print("Probability that every visitor buys a souvenir =",
prob_everyone)
print("Percentage =", prob_everyone * 100)

prob_max_7 = 0
probability_values = []

for x in range(11):
    probability = comb(n, x) * (p ** x) * (q ** (n - x))
    probability_values.append(probability)
    if x &lt;= 7:
        prob_max_7 += probability

print("\nQUESTION 2")
print("Probability that at most 7 visitors buy =", prob_max_7)
print("Percentage =", prob_max_7 * 100)

distribution = pd.DataFrame({
    "Visitors Buying": range(11),
    "Probability": probability_values
})

print("\nComplete Probability Distribution")
print(distribution)
print("\nTotal Probability =", sum(probability_values))

print("\nFINAL ANSWERS")
print("P(X=10) =", round(prob_everyone, 4))
print("P(X&lt;=7) =", round(prob_max_7, 4))

plt.figure(figsize=(8, 5))
plt.bar(range(11), probability_values)
plt.xlabel("Number of Visitors Buying Souvenirs")
plt.ylabel("Probability")
plt.title("Binomial Probability Distribution")
plt.xticks(range(11))
plt.grid(axis="y")
plt.show()
```

QUESTION 3

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\sat_score - SAT
Scores.csv")

print("First Five Records")
print(df.head())

print("\nColumn Names")
print(df.columns)

total_students = len(df)
print("\nTotal Students =", total_students)

less_800 = len(df[df["SAT_score"] &lt; 800])

more_1300 = len(df[df["SAT_score"] &gt; 1300])

prob_less_800 = less_800 / total_students
prob_more_1300 = more_1300 / total_students

print("\nStudents scoring below 800 =", less_800)
print("Probability =", prob_less_800)
print("Percentage =", prob_less_800 * 100)

print("\nStudents scoring above 1300 =", more_1300)
print("Probability =", prob_more_1300)
print("Percentage =", prob_more_1300 * 100)

plt.figure(figsize=(8, 5))
plt.hist(df["SAT_score"], bins=20)
plt.title("Distribution of SAT Scores")
plt.xlabel("SAT Score")
plt.ylabel("Number of Students")
plt.grid(True)
plt.show()
```

QUESTION 4

```
import pandas as pd
from scipy.stats import ttest_1samp

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\MobileTime -
Mobile Time.csv")

print("First Five Records")
print(df.head())

print("\nColumn Names")
print(df.columns)

n = len(df)
print("\nTotal Number of Observations =", n)

sample_mean = df["MobileTime"].mean()
print("\nSample Mean =", sample_mean)

population_mean = 144

t_statistic, p_value = ttest_1samp(df["MobileTime"],
population_mean)

print("\nOne-Sample t-Test Results")
print("t-statistic =", t_statistic)
print("p-value =", p_value)

alpha = 0.05
print("\nLevel of Significance =", alpha)

if p_value &lt; alpha:
    print("\nReject the Null Hypothesis")
    print("Conclusion: There is sufficient evidence that")
    print("the population mean mobile Internet usage time")
    print("is different from 144 minutes.")
else:
    print("\nFail to Reject the Null Hypothesis")
    print("Conclusion: There is not sufficient evidence that")
    print("the population mean mobile Internet usage time")
    print("is different from 144 minutes.")
```

QUESTION 5

```
import pandas as pd
from scipy.stats import ttest_ind

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\Luggage -
Luggage.csv")

print("First Five Records")
print(df.head())

print("\nColumn Names")
print(df.columns)

wing_a = df[df["Wing"] == "A"]["DeliveryTime"]

wing_b = df[df["Wing"] == "B"]["DeliveryTime"]

n_a = len(wing_a)
n_b = len(wing_b)
print("\nNumber of deliveries in Wing A =", n_a)
print("Number of deliveries in Wing B =", n_b)

mean_a = wing_a.mean()
mean_b = wing_b.mean()
print("\nMean Delivery Time - Wing A =", mean_a)
print("Mean Delivery Time - Wing B =", mean_b)

t_statistic, p_value = ttest_ind(wing_a, wing_b)

alpha = 0.05
print("\nIndependent Two-Sample t-Test Results")
print("t-statistic =", t_statistic)
print("p-value =", p_value)

print("\nLevel of Significance =", alpha)

if p_value &lt; alpha:
    print("\nReject the Null Hypothesis")
    print("Conclusion: There is a significant difference")
    print("between the mean delivery times of Wing A and Wing
B.")
else:
    print("\nFail to Reject the Null Hypothesis")
    print("Conclusion: There is no significant difference")
    print("between the mean delivery times of Wing A and Wing
B.")
```

QUESTION 6

```
import pandas as pd
from scipy.stats import ttest_rel

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\Concrete -
Concrete.csv")

print("First Five Records")
print(df.head())

print("\nColumn Names")
print(df.columns)

n = len(df)
print("\nTotal Number of Samples =", n)

strength_2_days = df["2_days"]

strength_7_days = df["7_days"]

mean_2_days = strength_2_days.mean()
mean_7_days = strength_7_days.mean()
print("\nMean Strength at 2 Days =", mean_2_days)
print("Mean Strength at 7 Days =", mean_7_days)

t_statistic, p_value = ttest_rel(strength_2_days,
strength_7_days)

alpha = 0.01

print("\nPaired t-Test Results")
print("t-statistic =", t_statistic)
print("p-value =", p_value)

print("\nLevel of Significance =", alpha)

if p_value &lt; alpha:
    print("\nReject the Null Hypothesis")
    print("Conclusion: There is sufficient evidence that")
    print("the mean concrete strength at 2 days is lower")
    print("than the mean concrete strength at 7 days.")
else:
    print("\nFail to Reject the Null Hypothesis")
    print("Conclusion: There is not sufficient evidence that")
    print("the mean concrete strength at 2 days is lower")
    print("than the mean concrete strength at 7 days.")
```

QUESTION 7

```
import pandas as pd
from scipy.stats import chi2_contingency

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\Promotion -
Promotions.csv")

print("Promotion Data")
print(df)

observed = [
    [15, 9],
    [16, 15]
]

chi_square, p_value, degrees_of_freedom, expected =
chi2_contingency(observed, correction=False)

print("\nExpected Frequencies")
print(expected)

print("\nChi-Square Test Results")
print("Chi-Square Statistic =", chi_square)
print("Degrees of Freedom =", degrees_of_freedom)
print("p-value =", p_value)

alpha = 0.05
print("\nLevel of Significance =", alpha)

if p_value &lt; alpha:
    print("\nReject the Null Hypothesis")
    print("Conclusion: There is a significant association")
    print("between company and promotion status.")
else:
    print("\nFail to Reject the Null Hypothesis")
    print("Conclusion: There is no significant association")
    print("between company and promotion status.")
```

QUESTION 8

```
import pandas as pd
from scipy.stats import f_oneway

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\carbonemissions
- Carbon Emissions.csv")

print("First Five Records")
print(df.head())

print("\nColumn Names")
print(df.columns)

total_observations = len(df)
print("\nTotal Number of Observations =", total_observations)

print("\nFuel Types")
print(df["FuelType"].unique())

print("\nMean Carbon Emission for Each Fuel Type")
mean_values = df.groupby("FuelType")["CarbonEmission"].mean()
print(mean_values)

petrol = df[df["FuelType"] == "Petrol"]["CarbonEmission"]
diesel = df[df["FuelType"] == "Diesel"]["CarbonEmission"]
hybrid = df[df["FuelType"] == "Hybrid"]["CarbonEmission"]
electric = df[df["FuelType"] == "Electric"]["CarbonEmission"]

f_statistic, p_value = f_oneway(petrol, diesel, hybrid,
electric)

alpha = 0.05

print("\nOne-Way ANOVA Results")
print("F-statistic =", f_statistic)
print("p-value =", p_value)

print("\nLevel of Significance =", alpha)

if p_value &lt; alpha:
    print("\nReject the Null Hypothesis")
    print("Conclusion: There is sufficient evidence")
    print("that mean carbon emissions differ")
    print("according to fuel type.")
else:
    print("\nFail to Reject the Null Hypothesis")
    print("Conclusion: There is not sufficient evidence")
    print("that mean carbon emissions differ")
    print("according to fuel type.")
```

QUESTION 9

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\Puneeth\Downloads\laptops -
Laptops.csv")

print("First Five Records")
print(df.head())

print("\nColumn Names")
print(df.columns)

total_observations = len(df)
print("\nTotal Number of Observations =", total_observations)

numeric_columns = df.select_dtypes(include="number").columns
print("\nNumerical Columns")
print(numeric_columns)

correlation_matrix = df[numeric_columns].corr()
print("\nCorrelation Matrix")
print(correlation_matrix)

price_correlation = correlation_matrix["Price"].sort_values(
ascending=False)
print("\nCorrelation with Laptop Price")
print(price_correlation)

print("\nFactors Related to Laptop Price")
for factor, correlation in price_correlation.items():
    print(factor, "=", round(correlation, 4))

other_factors = price_correlation.drop("Price")
strongest_factor = other_factors.abs().idxmax()
strongest_correlation = other_factors[strongest_factor]
print("\nStrongest Relationship with Price:")
print(strongest_factor, "=", round(strongest_correlation, 4))

plt.figure(figsize=(10, 6))
other_factors.sort_values().plot(kind="barh", color="skyblue",
 edgecolor="black")

plt.xlabel("Correlation Coefficient")
plt.ylabel("Factors")
plt.title("Correlation of Laptop Price with Other Factors")
plt.grid(axis="x", linestyle="--", alpha=0.7)
plt.show()
```